

CS23334 Fundamentals of Data Science Lab

Experiment 5

Data Collection and Initial Exploration

Objective: To collect, load, and perform initial exploration of the diabetes dataset.

Materials Needed:

- Diabetes dataset (available from sources such as UCI Machine Learning Repository or Kaggle)
- Python with libraries: pandas, numpy, matplotlib, seaborn

Procedure and Code:

1. Data Collection:

```
import pandas as pd

diabetes_df = pd.read_csv("diabetes.csv")

print(diabetes_df.head())
```

OUTPUT:

⇒

	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	BMI
0	6	148	72	35	0	33.6
1	1	85	66	29	0	26.6
2	8	183	64	0	0	23.3
3	1	89	66	23	94	28.1
4	0	137	40	35	168	43.1

	DiabetesPedigreeFunction	Age	Outcome
0	0.627	50	1
1	0.351	31	0
2	0.672	32	1
3	0.167	21	0
4	2.288	33	1

2. Initial Exploration:

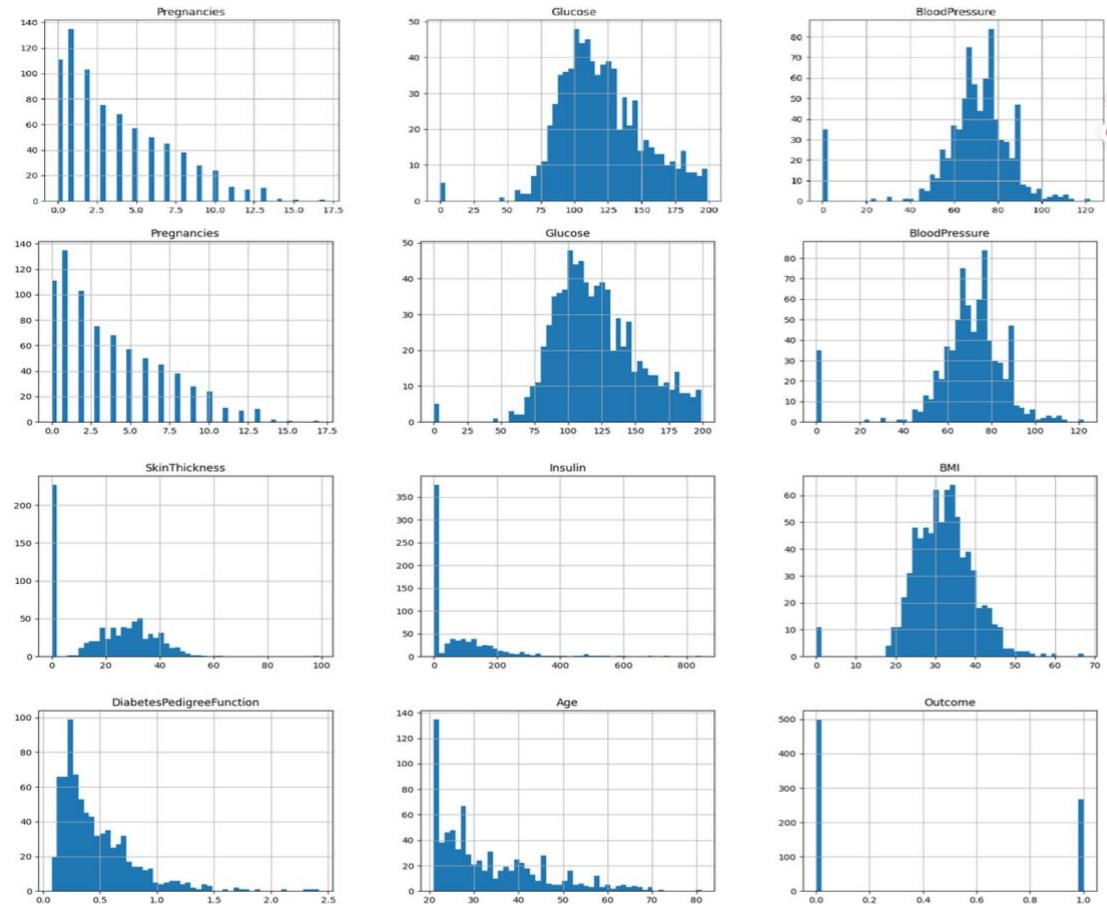
```
print(db.info())  
print(db.describe())  
  
import matplotlib.pyplot as plt  
import seaborn as sns  
  
db.hist(bins=50, figsize=(20, 15))  
plt.show()  
  
sns.pairplot(db)  
plt.show()
```

OUTPUT:

```
[3]
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 768 entries, 0 to 767
Data columns (total 9 columns):
#   Column                      Non-Null Count  Dtype
---  ---                      ---
0   Pregnancies                 768 non-null    int64
1   Glucose                     768 non-null    int64
2   BloodPressure               768 non-null    int64
3   SkinThickness               768 non-null    int64
4   Insulin                     768 non-null    int64
5   BMI                         768 non-null    float64
6   DiabetesPedigreeFunction    768 non-null    float64
7   Age                         768 non-null    int64
8   Outcome                     768 non-null    int64
dtypes: float64(2), int64(7)
memory usage: 54.1 KB
None
```

	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	\
count	768.000000	768.000000	768.000000	768.000000	768.000000	
mean	3.845052	120.894531	69.105469	20.536458	79.799479	
std	3.369578	31.972618	19.355807	15.952218	115.244002	
min	0.000000	0.000000	0.000000	0.000000	0.000000	
25%	1.000000	99.000000	62.000000	0.000000	0.000000	
50%	3.000000	117.000000	72.000000	23.000000	30.500000	
75%	6.000000	140.250000	80.000000	32.000000	127.250000	
max	17.000000	199.000000	122.000000	99.000000	846.000000	

OUTPUT:



OUTPUT:

